

Mamba4Rec with LLM Fusion

This is an extended version of [Mamba4Rec](#) that integrates LLM-based mood and intent signals for enhanced personalization.

Overview

The fusion architecture combines:

- **Mamba4Rec**: Sequential recommendation via Selective State Space Models
- **LLM Encoder**: Sentence transformer for mood/intent understanding
- **Gated Fusion**: Adaptive blending of sequential history and current mood

Installation

```
# Create conda environment
conda env create -f environment.yaml
conda activate mamba4rec-fusion

# Or install with pip
pip install torch mamba-ssm causal-conv1d recbole sentence-transformers
```

Training

Option 1: Phased Training (Recommended)

```
# Phase 1: Train vanilla Mamba4Rec
python train_phases.py --phase 1 --config config.yaml

# Phase 2: Train LLM projection + fusion
python train_phases.py --phase 2 --checkpoint checkpoints/mamba_phase1.

# Phase 3: Joint fine-tuning (optional)
python train_phases.py --phase 3 --checkpoint checkpoints/mamba_phase2.
```

Option 2: Simple Training

```
# Vanilla Mamba4Rec
python run.py

# With fusion enabled
python run.py --fusion
```

Inference

```

from inference import RecommendationEngine

# Load trained model
engine = RecommendationEngine.load("checkpoints/mamba_production.pt")

# Get mood-aware recommendations
result = engine.recommend(
    user_id=123,
    session_id="session-abc",
    item_history=[101, 205, 312, 89], # Previously watched items
    mood_text="I want something cozy from the early 2000s",
    top_k=10
)

# Print recommendations with explanation
print(engine.explain_recommendation(result))

```

Architecture

```

User Message → LLM Encoder → LLMPProjection → m_current
                                     ↓
User History → Mamba Layers → seq_output → [PreferenceFusion] → Recomme
                                     ↑
User Profile —————→ p_profile

```

Files

File	Description
mamba4rec.py	Core model with fusion integration
fusion.py	Gated fusion module
llm_projection.py	LLM to hidden space projection
llm_encoder.py	Sentence transformer wrapper
embedding_store.py	Mood and profile vector storage
train_phases.py	Phased training orchestrator
inference.py	High-level inference API
config.yaml	Configuration with fusion settings

Citation

```
@article{liu2024mamba4rec,  
  title={Mamba4rec: Towards efficient sequential recommendation with se  
  author={Liu, Chengkai and Lin, Jianghao and Wang, Jianling and Liu, H  
  journal={arXiv preprint arXiv:2403.03900},  
  year={2024}  
}
```